Task 3 – Data Visualization & Reporting

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1. Introduction

Human Resource Analytics plays a crucial role in enabling organizations to make data-driven workforce decisions. Employee attrition is one of the most critical challenges faced by HR departments, as high turnover leads to increased recruitment costs, reduced productivity, and loss of organizational knowledge.

The objective of this analysis is to transform statistical findings from previous exploratory analysis into meaningful visual insights that can support HR decision-making. Using the IBM HR Employee Attrition dataset, this report presents key trends related to attrition, salary distribution, job satisfaction, overtime, and departmental performance. The visualizations are designed to communicate insights clearly to both technical and non-technical HR stakeholders.

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2. Dataset Overview

The dataset used for this analysis is the IBM HR Employee Attrition dataset, containing detailed employee-level information.

Dataset Summary:

- Total Employees: 1470
- Number of Features: 35
- Target Variable: Attrition (Yes/No)

Key Variables Used:

- Attrition
- Department
- JobRole
- MonthlyIncome
- JobSatisfaction
- YearsAtCompany
- OverTime
- Age

The dataset contains demographic, job-related, compensation, and satisfaction metrics that help identify patterns influencing employee turnover.

There were no major missing values, and the dataset was cleaned and validated in Week 1 before conducting analysis.

3️⃣ Visualization Analysis

This section explains each visualization used in the dashboard and the insights derived from them.

🔹 3.1 KPI Overview

Key Performance Indicators calculated:

- Total Employees
- Attrition Rate
- Average Monthly Income
- Average Years at Company

Insight:

The attrition rate is approximately 16%, indicating that nearly 1 in 6 employees leave the organization. This is a significant turnover level and requires strategic HR intervention.

The average years at company suggests moderate employee tenure, meaning retention efforts could focus on mid-level employees.

🔹 3.2 Attrition by Department

A bar chart was used to visualize attrition across departments.

Insight:

The Sales department shows the highest attrition levels compared to Research & Development and Human Resources.

Business Interpretation:

Sales roles often involve high pressure, performance targets, and extended working hours. HR should investigate workload distribution, incentive structures, and managerial practices within the Sales department.

◈ 3.3 Attrition by Job Role

The visualization comparing job roles highlights which specific positions experience higher turnover.

Insight:

Sales Executives and Laboratory Technicians show relatively higher attrition counts.

Business Interpretation:

Role-specific dissatisfaction or limited growth opportunities may be influencing employees in these roles to leave. Career progression frameworks should be reviewed.

◈ 3.4 Monthly Income Distribution

A histogram was used to understand salary distribution.

Insight:

Most employees fall within the lower to mid-income range, with fewer employees in higher income brackets.

Business Interpretation:

Compensation imbalance may contribute to attrition, particularly if lower-income employees feel undervalued. A compensation benchmarking analysis is recommended.

◈ 3.5 Job Satisfaction vs Attrition

A boxplot was used to compare job satisfaction levels between employees who left and those who stayed.

Insight:

Employees who left tend to have lower job satisfaction levels compared to those who stayed.

Business Interpretation:

Job satisfaction is a strong predictor of attrition. Improving engagement initiatives, recognition programs, and managerial communication could help reduce turnover.

◈ 3.6 Overtime vs Attrition

A count plot was used to examine the relationship between overtime and attrition.

Insight:

Employees who work overtime show significantly higher attrition rates compared to those who do not.

Business Interpretation:

Work-life imbalance appears to be a major contributor to employee turnover. HR should evaluate workload distribution, overtime policies, and stress management practices.

4⃣ Correlation Summary

A correlation heatmap was generated to understand relationships between numerical variables.

Key Observations:

- Monthly Income is positively correlated with Years at Company and Job Level.
- Age is positively related to Years at Company.
- There is a weak negative relationship between Job Satisfaction and Attrition (when numerically encoded).

Interpretation:

Employees with longer tenure and higher job levels earn more, which is expected. However, satisfaction and overtime variables appear to influence attrition more strongly than demographic factors.

This indicates that behavioral and workplace environment factors may be stronger attrition drivers than age or income alone.

5⃣ HR Insights & Recommendations

Based on the visual and statistical analysis, the following recommendations are proposed:

1⃣ Reduce Overtime Burden

Implement workload balancing strategies and monitor overtime hours. Introduce flexible scheduling or additional staffing in high-pressure departments.

2⃣ Improve Job Satisfaction Programs

Conduct regular engagement surveys and introduce recognition programs to increase employee morale.

3⃣ Focus on Sales Department Retention

Develop targeted retention strategies for Sales teams, such as performance incentives, mentorship programs, and stress management initiatives.

4⃣ Compensation Benchmarking

Review salary structures for lower-income brackets to ensure market competitiveness.

5⃣ Career Development Initiatives

Establish clear promotion pathways and skill development programs to reduce turnover among mid-level employees.

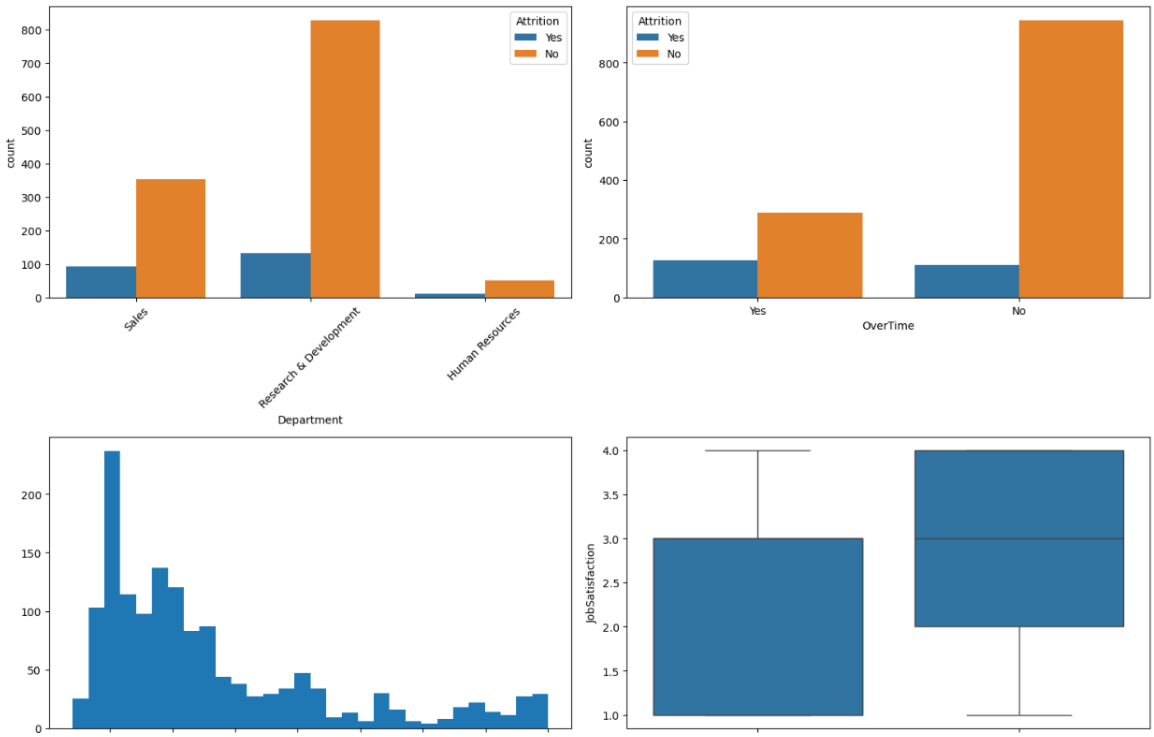
6⃣ Conclusion

This HR analytics study demonstrates how data visualization can transform raw employee data into actionable business insights.

The analysis reveals that attrition is influenced primarily by overtime workload, job satisfaction levels, and departmental pressure, particularly within Sales roles. While compensation and tenure play supporting roles, employee engagement and work-life balance appear to be stronger determinants of turnover.

By implementing targeted retention strategies based on these findings, the organization can reduce attrition, improve employee satisfaction, and enhance long-term workforce stability.

This project highlights the importance of data-driven HR decision-making and the value of combining analytical techniques with clear visual storytelling.




```
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber	Envi
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	
mean	36.923810	802.485714	9.192517	2.912925	1.0	1024.865306	
std	9.135373	403.509100	8.106864	1.024165	0.0	602.024335	
min	18.000000	102.000000	1.000000	1.000000	1.0	1.000000	
25%	30.000000	465.000000	2.000000	2.000000	1.0	491.250000	
50%	36.000000	802.000000	7.000000	3.000000	1.0	1020.500000	
75%	43.000000	1157.000000	14.000000	4.000000	1.0	1555.750000	
max	60.000000	1499.000000	29.000000	5.000000	1.0	2068.000000	

8 rows × 26 columns

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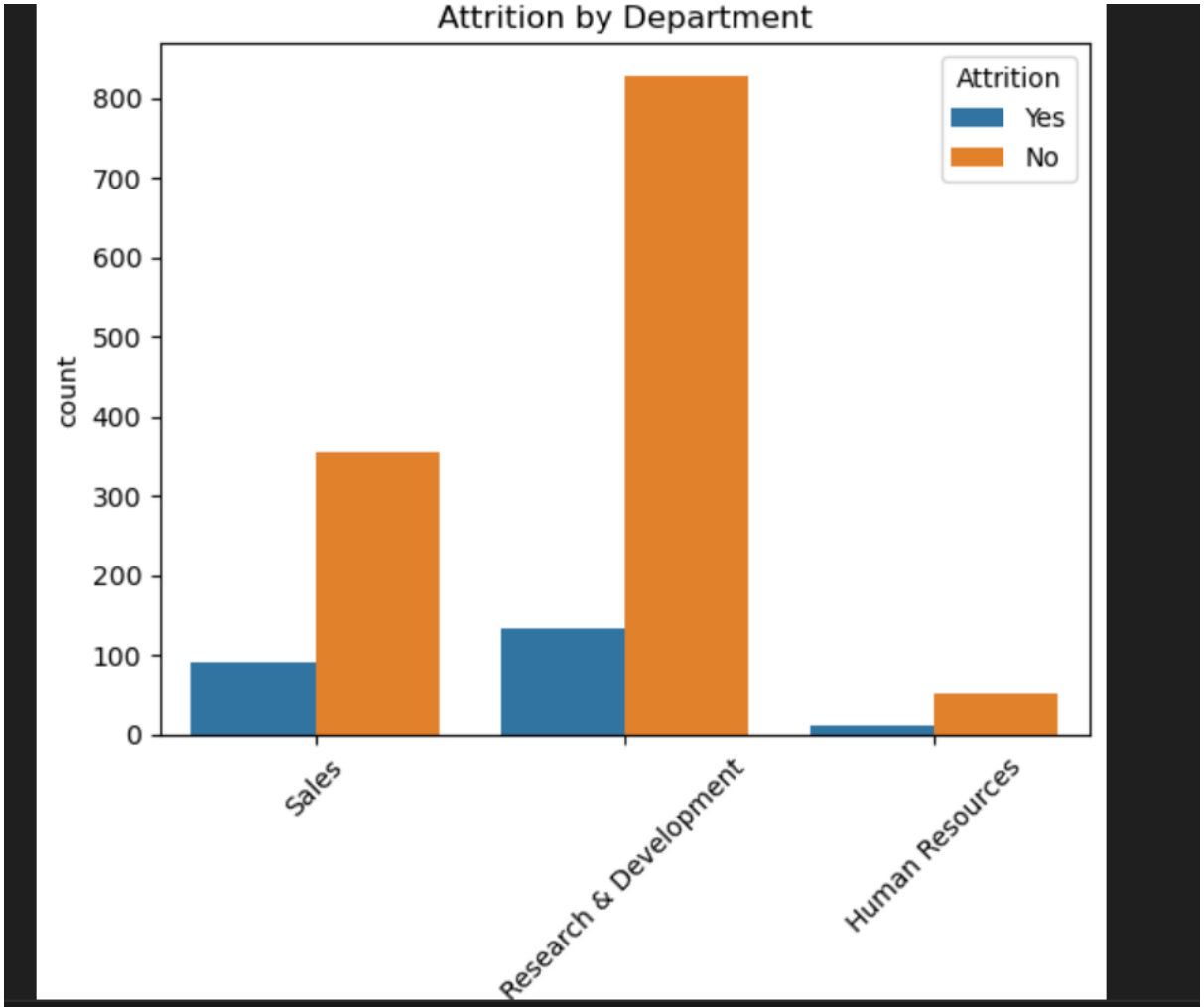
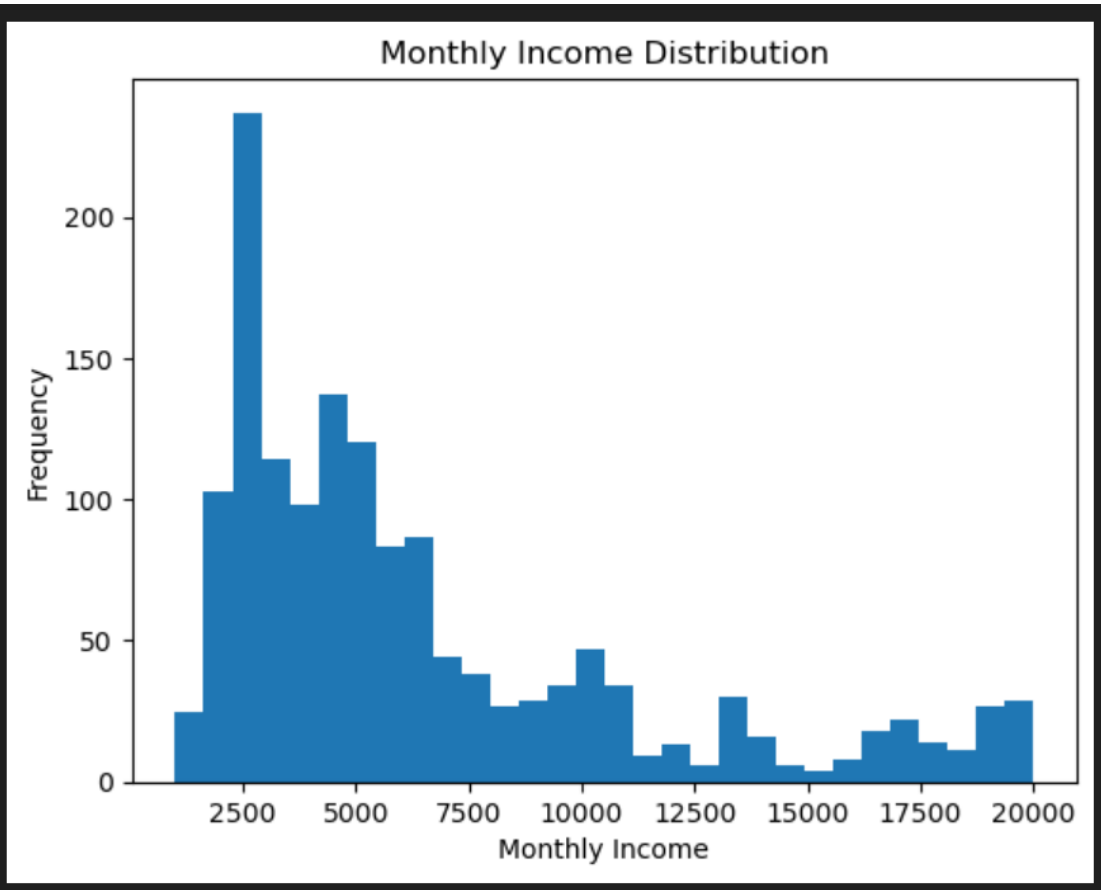
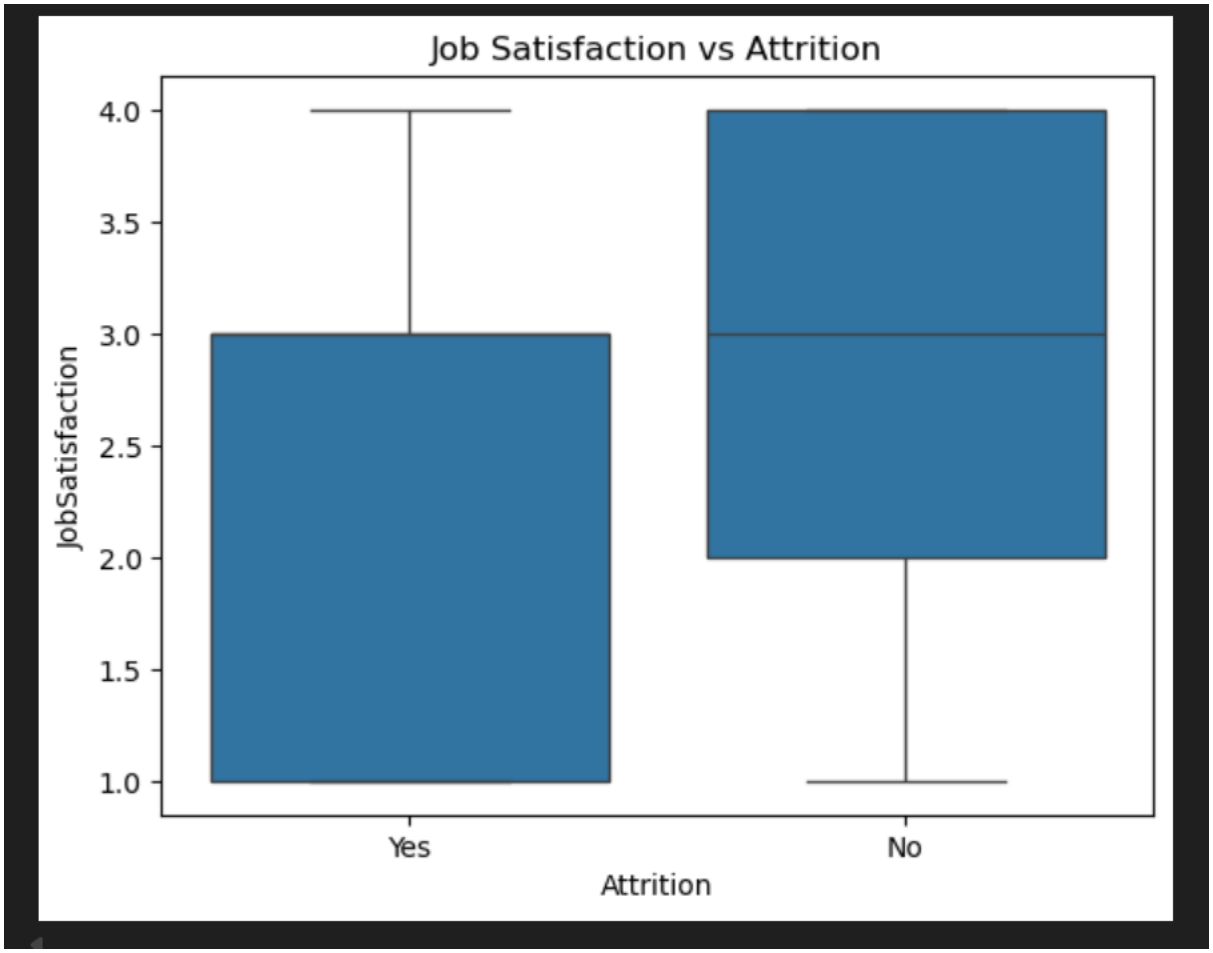
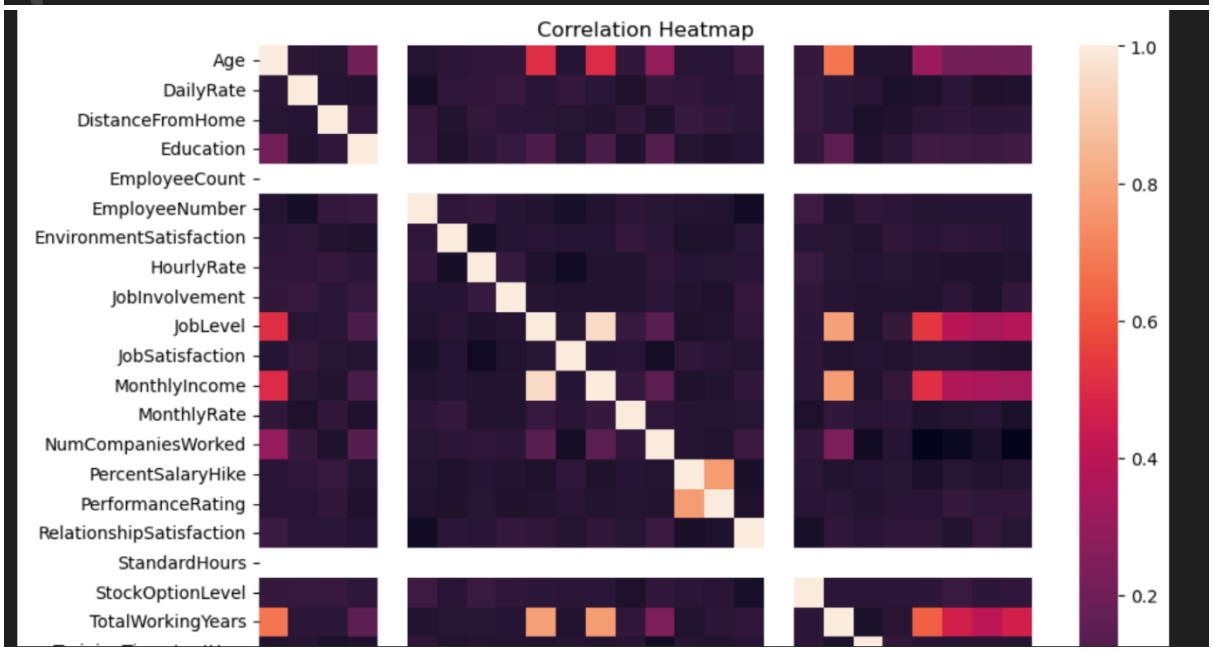
Total Employees: 1470
Attrition Rate: 16.12 %
Average Monthly Income: 6502.93
Average Years at Company: 7.01
```


Attrition by Department
Attrition
Yes
No
count
800
700
600
500
400
300
200
100
0
Sales
Research & Development
Human Resources


Monthly Income Distribution
Frequency
0
50
100
150
200
2500
5000
7500
10000
12500
15000
17500
20000
Monthly Income


Job Satisfaction vs Attrition
JobSatisfaction
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Yes
No
Attrition


Correlation Heatmap
Age
DailyRate
DistanceFromHome
Education
EmployeeCount
EmployeeNumber
EnvironmentSatisfaction
HourlyRate
JobInvolvement
JobLevel
JobSatisfaction
MonthlyIncome
MonthlyRate
NumCompaniesWorked
PercentSalaryHike
PerformanceRating
RelationshipSatisfaction
StandardHours
StockOptionLevel
TotalWorkingYears
1.0
0.8
0.6
0.4
0.2